Cevdet Batuhan Isik

Boston, MA | (617) 595-2314 | isik.c@northeastern.edu | linkedin.com/in/batuhan-isik32 | github.com/batuhanisik751 | batuhanisik.vercel.app

Education

Northeastern University, Khoury College of Computer Sciences — Boston, MA
B.S. in Computer Science, Concentration in Artificial Intelligence — *Expected December 2026*

Relevant Coursework: Machine Learning & Data Mining, Artificial Intelligence, Information Retrieval, Algorithms & Data, Object-Oriented Design, Computer Systems, Foundations of Data Science, Databases
Honors & Leadership: Dean's List (Spring 2023); Co-President, Turkish Student Association (TSA); Delta Kappa Epsilon Fraternity

Experience

Burnes Center for Social Change (InnovateUS) — January 2026 – June 2026
Generative AI Product Development Fellow — *Boston, MA*

- Built an **agentic, multi-agent RAG** coaching tool powered by **OpenAI LLMs** for **InnovateUS**, a national learning platform serving **200,000+** public-sector professionals; an orchestrator routes each user turn to one of **5** specialized prompt agents over a cited Weaviate knowledge base.
- Developed **full-stack** features across a **React/TypeScript frontend** (Tailwind, Cognito sign-in, accessibility and mobile fixes) and a **Node.js backend** of **REST APIs** on AWS Lambda, streaming chat responses over WebSockets.
- Designed the serverless **AWS** architecture in **CDK** (API Gateway, Lambda, Step Functions, DynamoDB, S3), including **Python** ingestion scripts that load documents into Weaviate, interview ingestion pipelines with failure handling, PII redaction via AWS Comprehend, and CloudWatch alarms on error rate and latency.
- Shipped through a **GitHub Actions CI/CD** pipeline running **Vitest** tests on every push to main; wrote the LLM output-quality and evaluation criteria and refined prompts based on errors we found.
- Conducted qualitative and quantitative user research with product managers to define requirements, managed the product lifecycle from problem statement to deployment, and communicated outcomes to technical and non-technical audiences.

OBSS Technology — June 2025 – August 2025
Deep Learning Intern — *Istanbul, Turkey*

- Engineered diffusion-based inpainting workflows in **ComfyUI** that automatically remove objects from images and regenerate realistic backgrounds.
- Built an image-realism evaluation algorithm in **FiftyOne** using **LPIPS** and **SSIM**, enabling objective benchmarking of generative model outputs against real images.

Northeastern University — July 2024 – December 2024
Skillstack Administrator Support Co-op — *Boston, MA*

- Audited and optimized a B2B **Canvas LMS** for Northeastern's non-credit certification programs, maintaining user reporting and sub-account structure.
- Ran office hours for professors, resolving Canvas issues live, and refined the support process and documentation for Canvas and **Canvas Credentials**.
- Built a **Salesforce** workflow automation prototype with Flow Builder and Einstein Copilot at a hackathon after completing 56 hours of admin training.

Projects

Quantifying Happiness | *Swift, SwiftUI, HealthKit, Python* — December 2025

- Built an iOS app that models how **9** HealthKit health metrics relate to user satisfaction and recommends daily changes.
- Applied simulated annealing over **KNN regression**, raising predicted satisfaction from **5.0 to 8.98** in 1,000 iterations with convergence toward the global optimum; validated end to end on a 100-record synthetic dataset.

Yelp Restaurant Retrieval System | *Python, SBERT, FAISS* — April 2025

- Built a location-aware search over **50,000+** Yelp reviews with SBERT embeddings and FAISS dense retrieval, re-ranked by rating and user proximity.
- Evaluated ranking quality against ground-truth labels with custom relevance tiers using **MAP@100**.

Hexagonal Reversi | *Java, JavaFX, JUnit* — December 2023, extended March 2026

- Built a complete Reversi game on a hexagonal board with an **MVC** architecture; migrated the GUI from Swing to **JavaFX** with keyboard navigation and move highlighting.
- Added AI opponents using **minimax with alpha-beta pruning**, save/load, and LAN multiplayer, backed by **JUnit** tests.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, Swift, C, SQL, R, HTML/CSS
Frameworks: React, Next.js, Node.js, Express, FastAPI, Tailwind CSS, SwiftUI, JavaFX
AI/ML: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, LLMs, LangChain, LlamaIndex, OpenAI API, RAG, Agentic & Multi-Agent Systems, Prompt Engineering, LLM Fine-Tuning, SBERT, FAISS
Cloud & DevOps: AWS (Lambda, API Gateway, Step Functions, DynamoDB, S3, EC2, Cognito, CloudWatch, Comprehend), CDK, Terraform, Docker, GitHub Actions
Databases: PostgreSQL, Supabase, DynamoDB, MongoDB, Redis, Weaviate
APIs & Concepts: REST APIs, GraphQL, WebSockets, Data Structures & Algorithms, OOP, Design Patterns, MVC, CI/CD
Testing & Tools: Linux, Git, Vitest, React Testing Library, JUnit, Postman, Jupyter, Claude Code, Cursor, GitHub Copilot, Figma